The New Tokenomics: How Data Infrastructure Increasingly Sets AI ROI

Five Ways SSDs Are Optimizing AI Outcomes and Efficiency

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Two things matter in every AI use case.

a. How valuable is the output?
b. What does it cost to obtain?

If a > b, do more of it.

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Over time, new AI capabilities add value but increasingly create or use more data, raising memory and storage requirements.

Training

(fixed costs)

- Bigger data sets
- Richer content types
- Longer training runs
- More frequent checkpoints

Inference

(variable costs)

- Longer context windows
- More complex reasoning
- Agentic loops and tool calls
- Greater persistence

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Therefore, the economics of every token are increasingly decided below the GPU, in data memory and storage.

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One task, five data infrastructure consequences.

A sports analytics question frames five steps, each backed by a published result.

"How did our red-zone defense change across these games?"



01

RETRIEVE

DiskANN for RAG

02

REUSE

Video DiskANN × KV cache

03

MOVE

KV cache with GDS

04

SHARE

Shared-pool KV cache

05

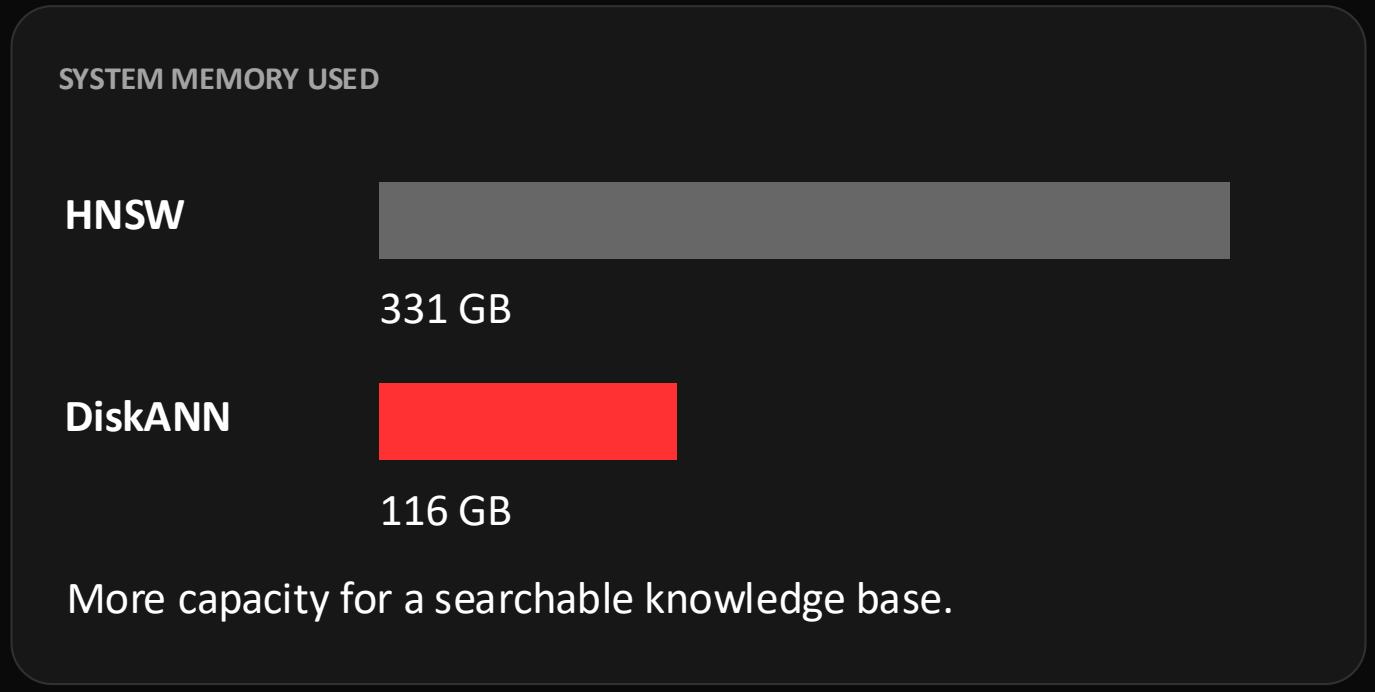
COMPRESS

TurboQuant

Sources: Solidigm and Metrum AI white paper (February 2026); Metrum AI blog posts (March and September 2026).

Search more data without scaling DRAM.

100 million vectors / HNSW in memory versus DiskANN on Solidigm D7-PS1010



65%

less DRAM

331 → 116 GB

+26%

query throughput

131 → 165 QPS

99.2%

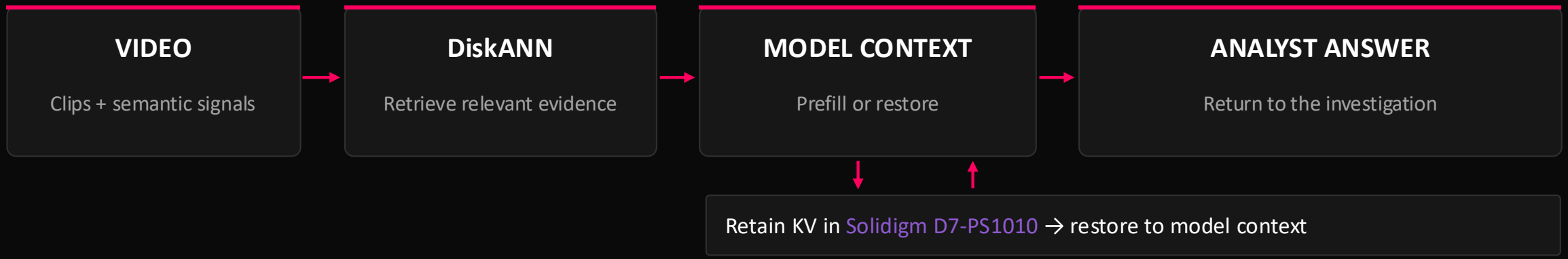
Recall@10

HNSW: 98.4%

Source: White paper “Breaking Memory Barriers in Video Analytics” by Solidigm and Metrum AI, February 2026 (LAION 100M).

Find evidence faster. Skip the recompute.

Two complementary roles for Solidigm SSDs: searchable video indexes and reusable model state.



VIDEO RETRIEVAL

+14% QPS

2,071 → 2,371 / 1,024 query threads

SPORTS CONTEXT REVISIT

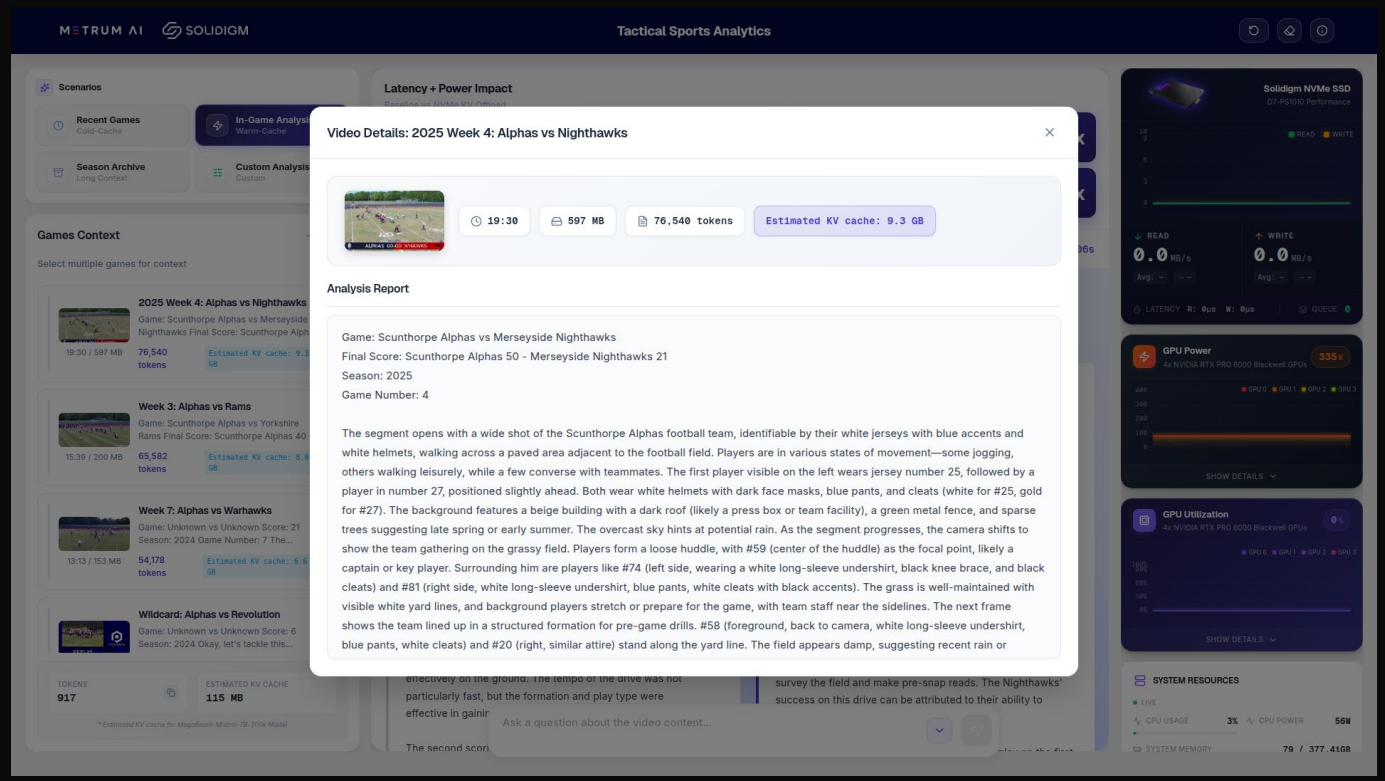
31 s → 2.8 s

Time to first token / Season Archive, single user

Sources: “Breaking Memory Barriers in Video Analytics,” Solidigm and Metrum AI, February 2026; “Eliminate Redundant GPU Compute,” Metrum AI, March 2026. Separate results, not a combined speedup.

Quick demo: From game film to a faster follow-up.

Select the footage, ask a question, get answers fast. Don’t recompute.



01 SELECT

Choose the game evidence.

02 ASK

Run the same question on both paths.

03 RETURN

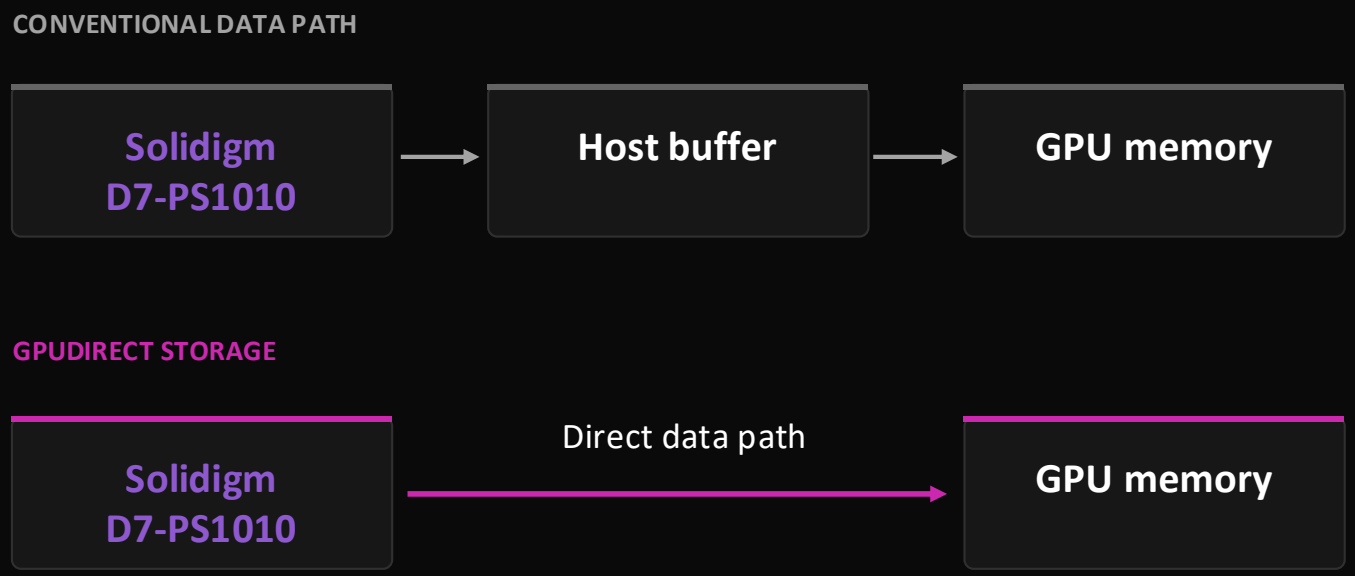
Show retained context being reused.

Source: Blog post “Eliminate Redundant GPU Compute” by Metrum AI, March 2026.

PRELIMINARY TESTING

More efficient KV cache reads with GPUDirect Storage.

NVIDIA GPUDirect Storage removes the host-memory bounce from a supported NVMe-to-GPU path.



CPU still orchestrates I/O.

100K-TOKEN REPEAT REQUESTS

4.95 → 1.73 s

Median end-to-end latency

9.41 → 1.85 GB/s

Host-memory traffic

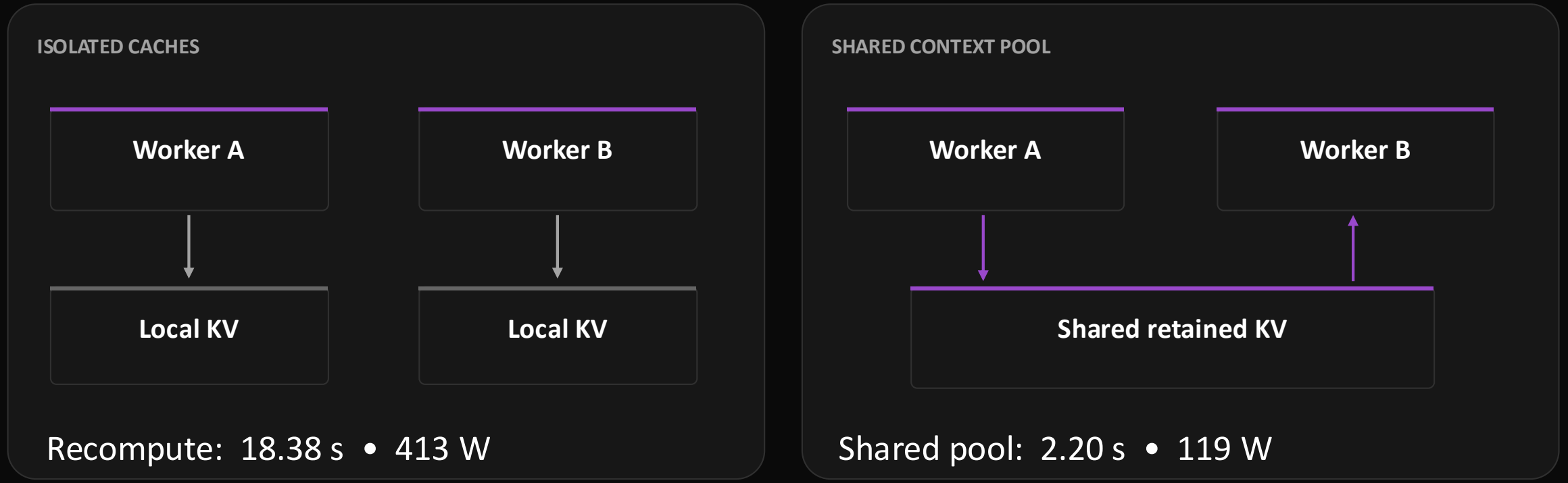
Same drive; a simpler path to the GPU.

Source: Blog post “What Happens When the KV Cache Skips Host Memory?” by Metrum AI, September 2026. Chunk sizes differ between the two paths.

PRELIMINARY TESTING

A shared SSD pool turns recompute into a read.

Round-robin across four replicas: each follow-up hits a replica that never saw the prompt.



Source: Blog post “Can a Shared SSD Pool Make KV Cache Reuse Work Across a Fleet?” by Metrum AI, September 2026. p50 time to first token and mean GPU power; 99K-token prompts, concurrency 1.

METRUM AI

One prefill, reused by any worker.

Coding agents: 6.7× faster first token from a shared Solidigm D7-PS1010 tier vs GPU memory.

01

PREPARE ON A

Warm one compatible context on worker A.

02

REQUEST ON B

Send the same-prefix follow-up to a second worker.

03

SHOW THE DIFFERENCE

Compare time to first token against GPU memory only.

RETURN VISIT

vs GPU memory only

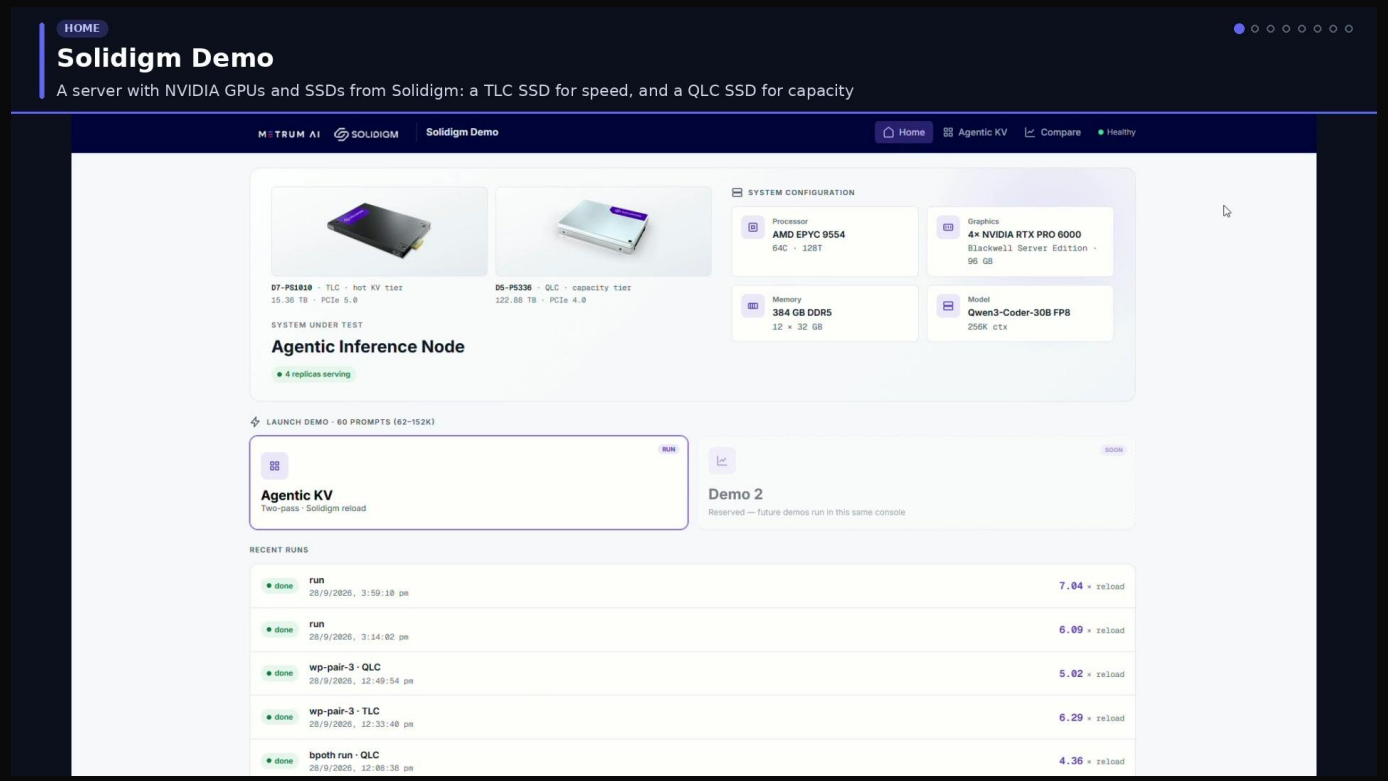
Solidigm D7-PS1010 6.7× faster first token (2.05 s vs 13.76 s)

Solidigm D5-P5336 4.4× faster first token (3.10 s vs 13.76 s)

Source: Metrum AI internal benchmarking, October 2026; blog post yet to be published. Return visit vs GPU memory only; 60 coding tasks, concurrency 4.

Quick demo: Shared KV on Solidigm, side by side.

Same 60 coding prompts: GPU memory alone vs a shared Solidigm SSD tier.



01 SET UP

Choose where the KV cache lives.

02 RUN

Compute and save, then reload.

03 COMPARE

GPU memory alone vs TLC and QLC SSDs.

Source: Metrum AI internal benchmarking, October 2026; blog post yet to be published.

METRUM AI

PRELIMINARY TESTING

Compression makes space for more context data.

Compress at the storage boundary: 9% decode cost vs 29% for the same codec in GPU memory.

3.8×

KV compression

4-bit codec versus BF16 KV: about 41K → 157K 30K-token sessions on one 122.88 TB Solidigm D5-P5336, by codec ratio.



4-bit matched the control; the tested 3-bit preset did not.

Source: Blog post “What Happens When Compression Moves to the SSD?” by Metrum AI, September 2026. Single configuration; quality at this test’s resolution. Restore at 4-bit added 9.6% warm-restore latency and 2.5× host CPU.

Data is everywhere in the AI pipeline.

Whether you are finding
reusing
moving
sharing
or shrinking it,

Solidigm SSDs can improve outcomes and efficiency.

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Every token has a resource cost.
Every resource has a dollar cost.

Spend less, go further with SSDs.

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Read the research behind every number.

Scan for the full write-ups: setups, methods and caveats.

ALL RESEARCH
metrum.ai/resource

01
RETRIEVE

Breaking Memory Barriers
White paper, February 2026

02
REUSE

Eliminate Redundant GPU Compute
Blog, March 2026

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KV cache skips host memory
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Shared SSD pool across a fleet
Blog, September 2026

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COMPRESS

Compression moves to the SSD
Blog, September 2026

Published results from this talk. The coding-agent results on slide 11 are internal and not yet published.



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